

PHIL Robot Software Quick User Guide (ANT)

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Version: 1

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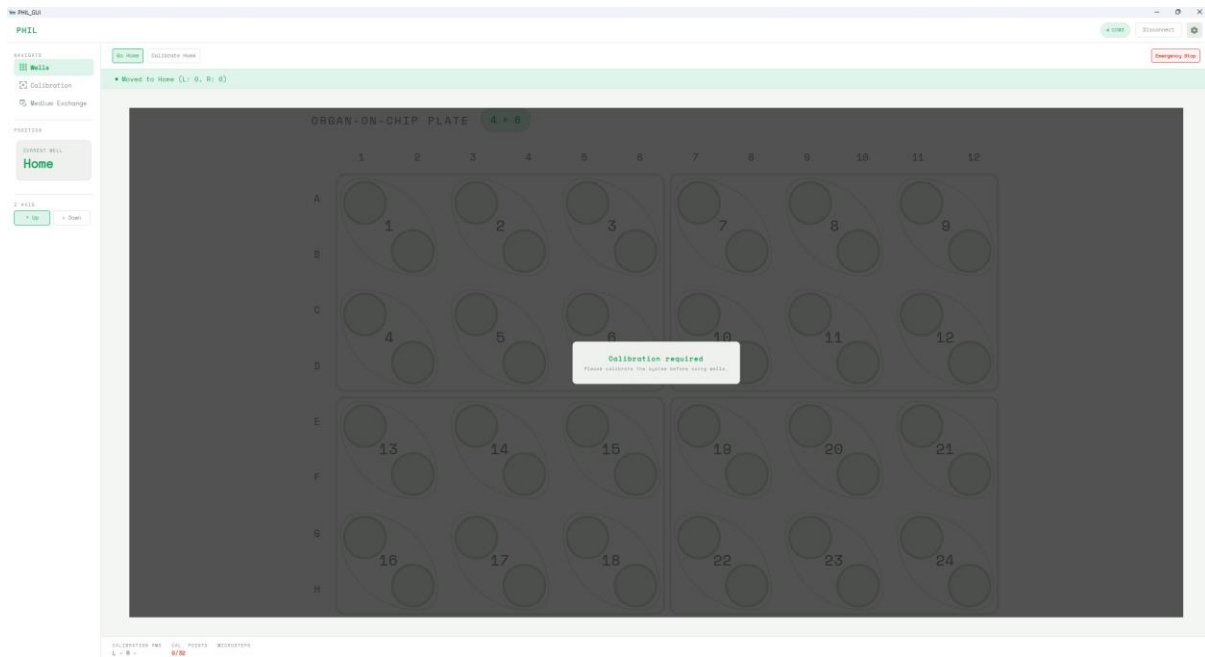
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1 Introduction

This handbook provides a simple overview of how to use the PHIL software. It is intended for first-time users and as a quick reference when assistance is needed. The software allows you to control the robot, perform calibration, manually operate pumps, and automate liquid handling through actions.

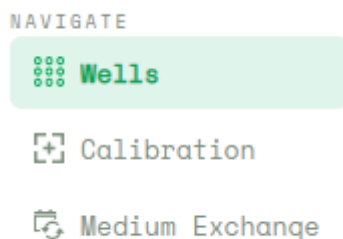
1.1 Main Window

The Main Window is the central interface of the application.

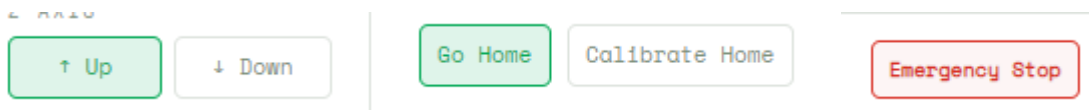


It contains:

- Sidebar → Navigate between different views

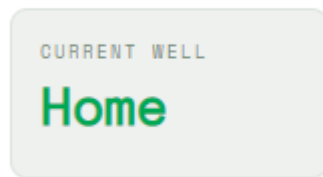


- Global controls → Z-axis movement, Homing, Emergency Stop

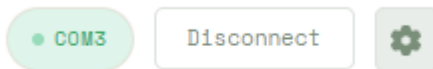


- Robot information → Current well, X/Y position, motor angles

POSITION



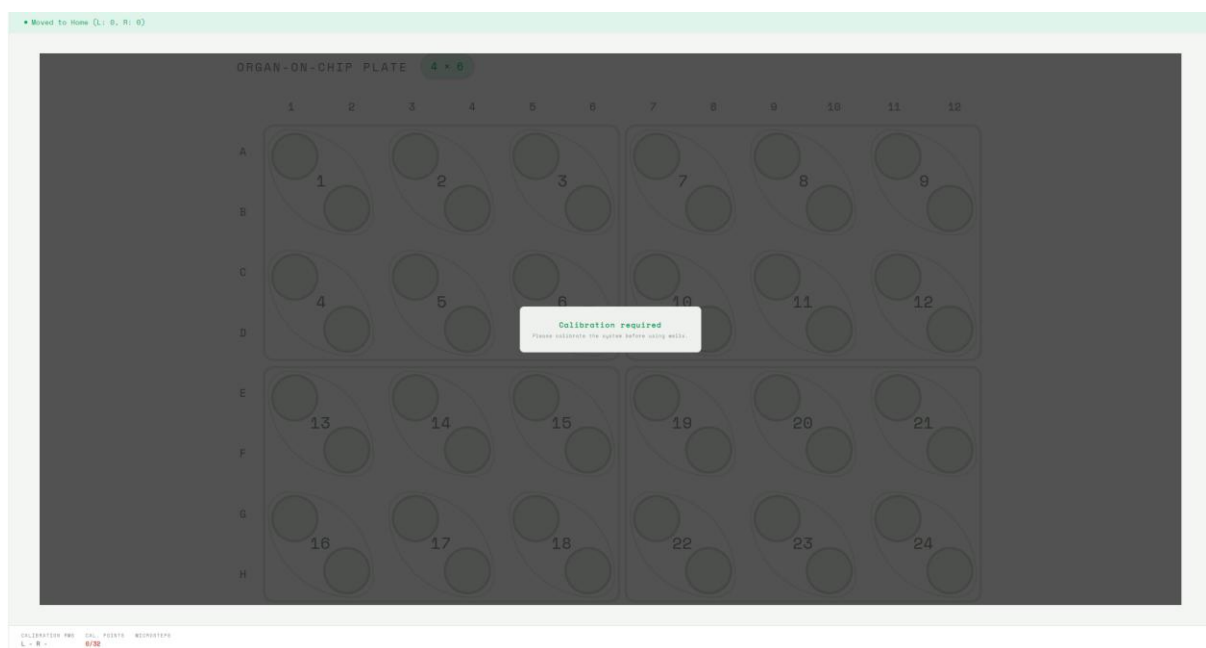
- Settings button → Open settings menu



- Disconnect button → Disconnect from the robot

1.2 Wells View

The Wells View is used to test calibrated positions.



- Access is restricted until calibration is completed
- Clicking a well moves the robot to that position

Top bar shows:

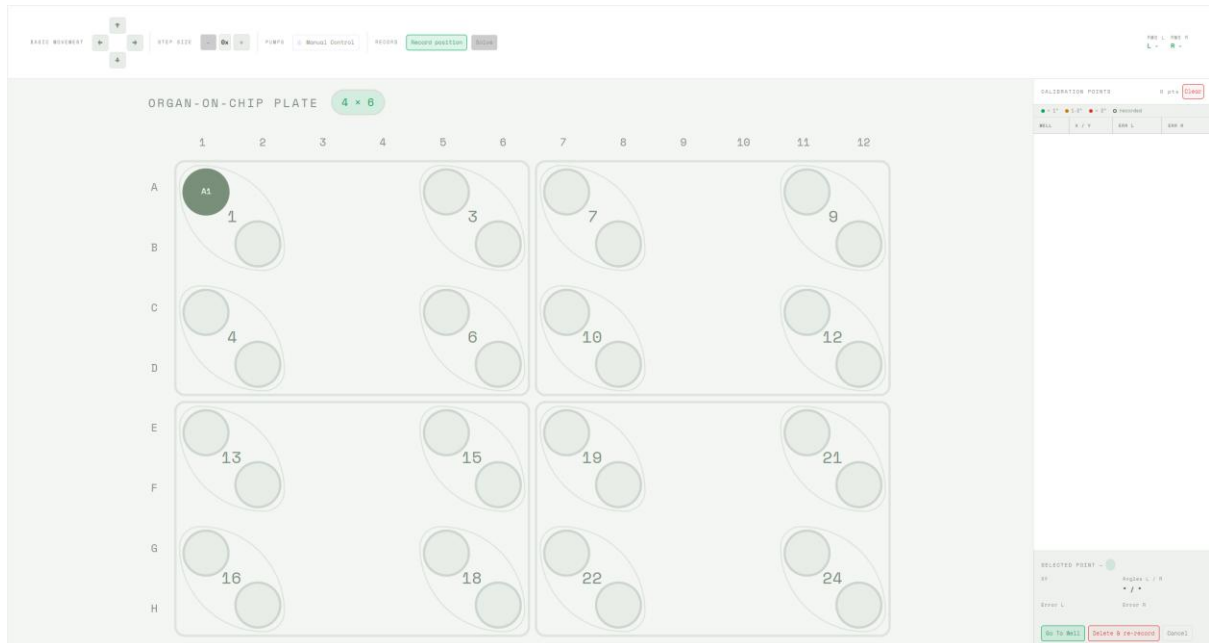
- Robot state (Moving / Idle / Moved) & Left/Right motor angles

Bottom bar shows:

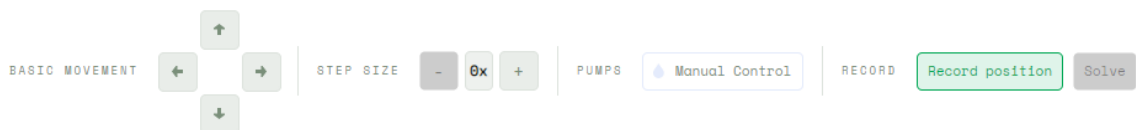
- Calibration RMS → average positioning error (lower is better)
- Number of calibration points
- Current microstep setting

1.3 Calibration View

The Calibration View is used to calibrate the well plate. A minimum of 10 points is needed.



The top bar offers handy controls.



- Basic movement controls (Forward, Backward, Left, Right)
- Step Size is used to adjust movement precision.
- Pump Control button opens Pumps Window.
- Record adds the current position to calibration list of points.
- Solve Map computes final calibration

When selecting a well → side panel opens



Here, you can see your selected point (if existent), the XY and Angles of L/R and Error Left and Error Right (if recorded).

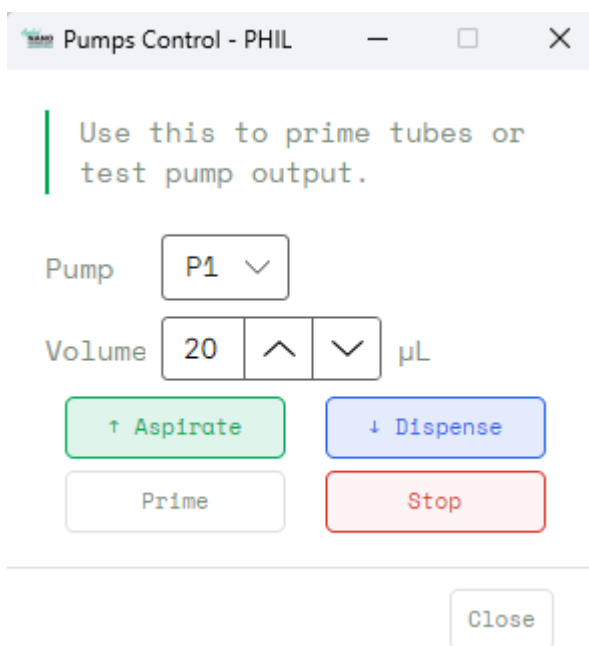
Moreover, you have the option to Go to well, Delete & re-record or cancel which will close the bottom menu and unselect your well.

If one point is incorrect you may select the point, go to well, manually position the pipette correctly with basic movement and then click on Delete & re-record. However, if all points are incorrect, you may Clear them.



1.4 Pumps Window

Accessed from the Calibration top bar, the Pumps Window allows manual pump control.

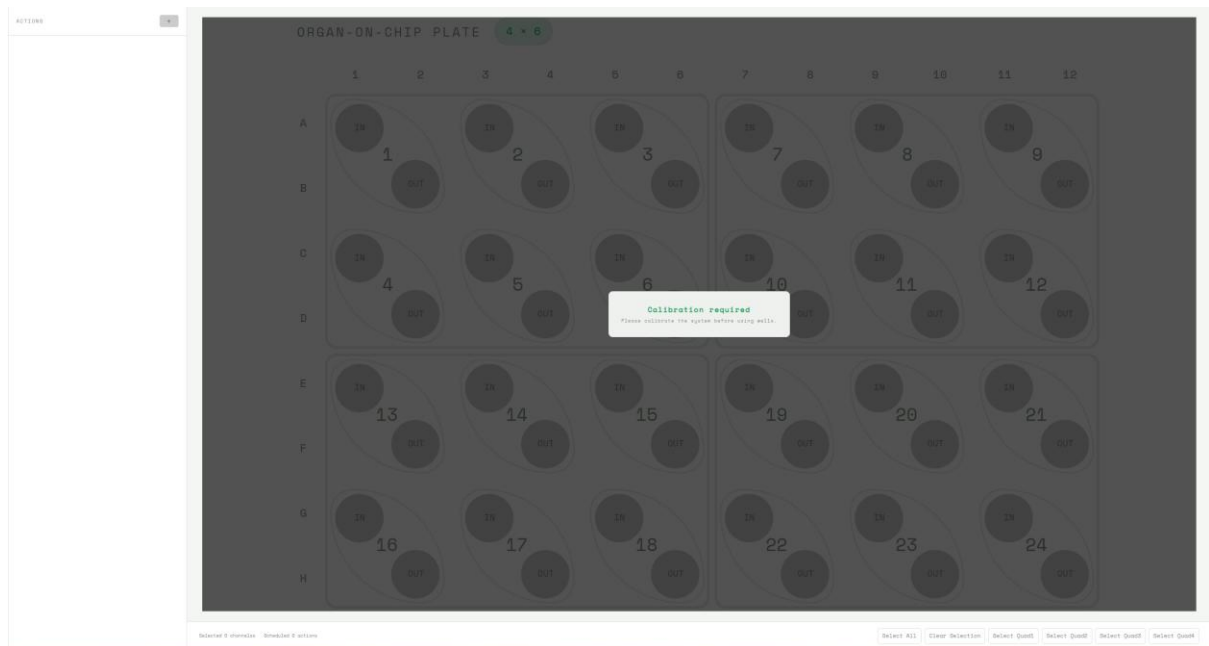


Here you can select your pump and either Aspirate or Dispense the specified Volume.

Prime is used when you wish to fill the empty tubing with liquid. The robot does not know when to stop, so when you see liquid exiting the pipette, you may press Stop.

1.5 Medium Exchange View

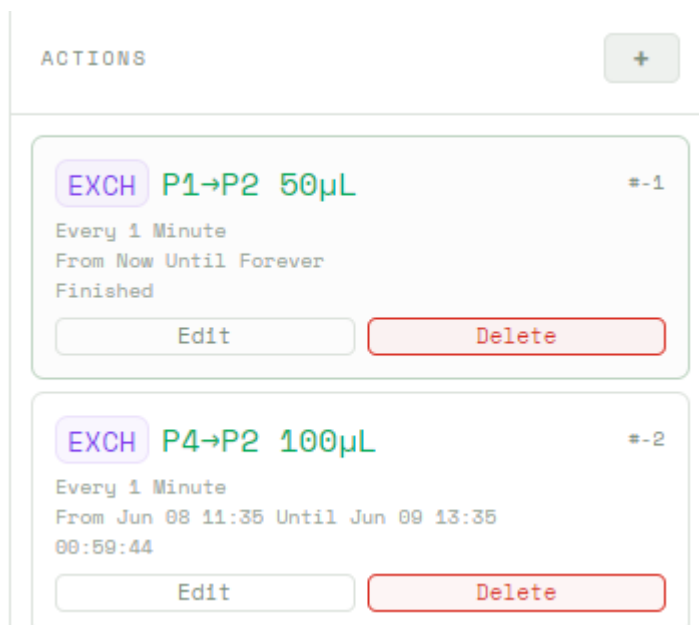
Used to create and manage automated actions.



On Medium Exchange View, you can Create, Manage, Attach or Detach Actions. Similarly to Wells View, if no calibration is available, you cannot access the well plate.

To create an Action, click on the + button on the top left sidebar. This will open the Action Window.

Once Created, Actions can be seen on the left sidebar.



- Edit will open the same Action Window but with your Action's information.
- Delete will Delete and Detach your Action from all your Channel(s) or Well(s).

ACTIONS

EXCH P1-P2 50µL

Every 1 Minute

From Now Until Forever

Finished

Edit

Delete

EXCH P4-P2 100µL

Every 1 Minute

From Jun 08 11:35 Until Jun 09 13:35

00:50:27

Edit

Delete

ORGAN-ON-CHIP PLATE 4 × 6

	1	2	3	4	5	6	7	8	9	10	11	12
A	IN 1	IN 2	IN 3	IN 4	IN 5	IN 6	IN 7	IN 8	IN 9	IN 10	IN 11	IN 12
B	OUT 1	OUT 2	OUT 3	OUT 4	OUT 5	OUT 6	OUT 7	OUT 8	OUT 9	OUT 10	OUT 11	OUT 12
C	IN 4	IN 5	IN 6	IN 7	IN 8	IN 9	IN 10	IN 11	IN 12	IN 13	IN 14	IN 15
D	OUT 4	OUT 5	OUT 6	OUT 7	OUT 8	OUT 9	OUT 10	OUT 11	OUT 12	OUT 13	OUT 14	OUT 15
E	IN 13	IN 14	IN 15	IN 16	IN 17	IN 18	IN 19	IN 20	IN 21	IN 22	IN 23	IN 24
F	OUT 13	OUT 14	OUT 15	OUT 16	OUT 17	OUT 18	OUT 19	OUT 20	OUT 21	OUT 22	OUT 23	OUT 24
G	IN 16	IN 17	IN 18	IN 19	IN 20	IN 21	IN 22	IN 23	IN 24	IN 25	IN 26	IN 27
H	OUT 16	OUT 17	OUT 18	OUT 19	OUT 20	OUT 21	OUT 22	OUT 23	OUT 24	OUT 25	OUT 26	OUT 27

Selected 1 channels Scheduled 0 actions

Select All

Clear Selection

Select Quad1

Select Quad2

Select Quad3

Select Quad4

WELL DETAILS

Channel 1

0 actions attached

ATTACHED

AVAILABLE TO ATTACH

EXCH P1-P2 50µL

+

EXCH P4-P2 100µL

+

Selecting a Channel/Well will open the right sidebar, where you can either Attach or Detach the Actions.

This sidebar provides the Actions that are existent in the current Well/Channel. If multiple Wells/Channels are selected, we will only see the Action in common.

You may select multiple Channels/Wells to either attach or detach Action(s).

1.6 Action Window

Action Window can be opened from either Creating an Action or Editing an Action.

New Action - PHIL

PUMP IN

P1

PUMP OUT

P2

AMOUNT (μL)

50

^

v

Repeatable

☒ Repeat

EVERY

1

^

v

UNIT

Minute

v

Schedule

START DATE & TIME ⓘ

day

month

hour

minute

×

END DATE & TIME ⓘ

day

month

hour

minute

×

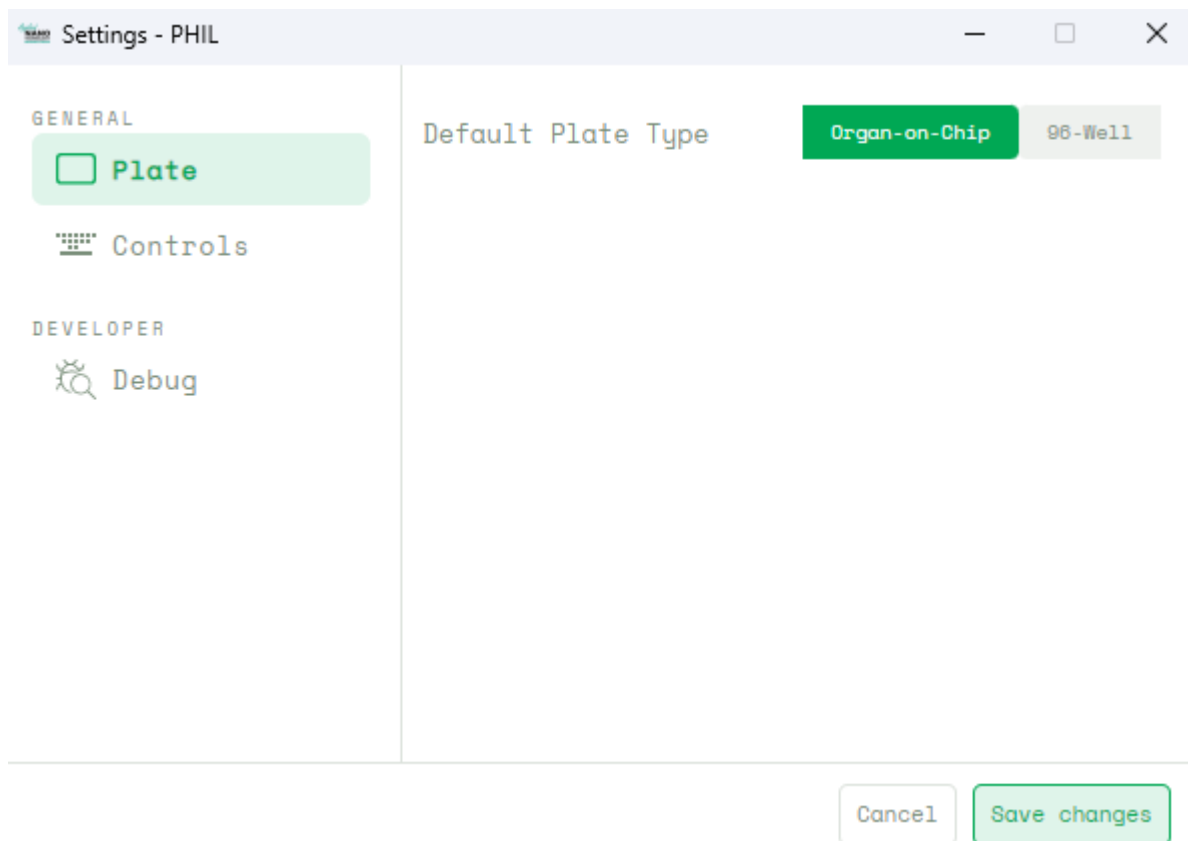
Cancel

Create Action

Here, select your Pump IN and Pump OUT. Pump IN will be assigned to the IN well of the Channel and will only Disperse the specified amount. Pump OUT will be assigned to the OUT well of the Channel and only Aspirate the specified amount. Pump IN and OUT are optional. If you wish for the Action to only execute once, uncheck the Repeat checkbox and select your Date and Time. Otherwise fill in both Start Date Time and End Date Time. If both are left empty the Action will start after the specified interval of time and run forever.

1.7 Settings Window

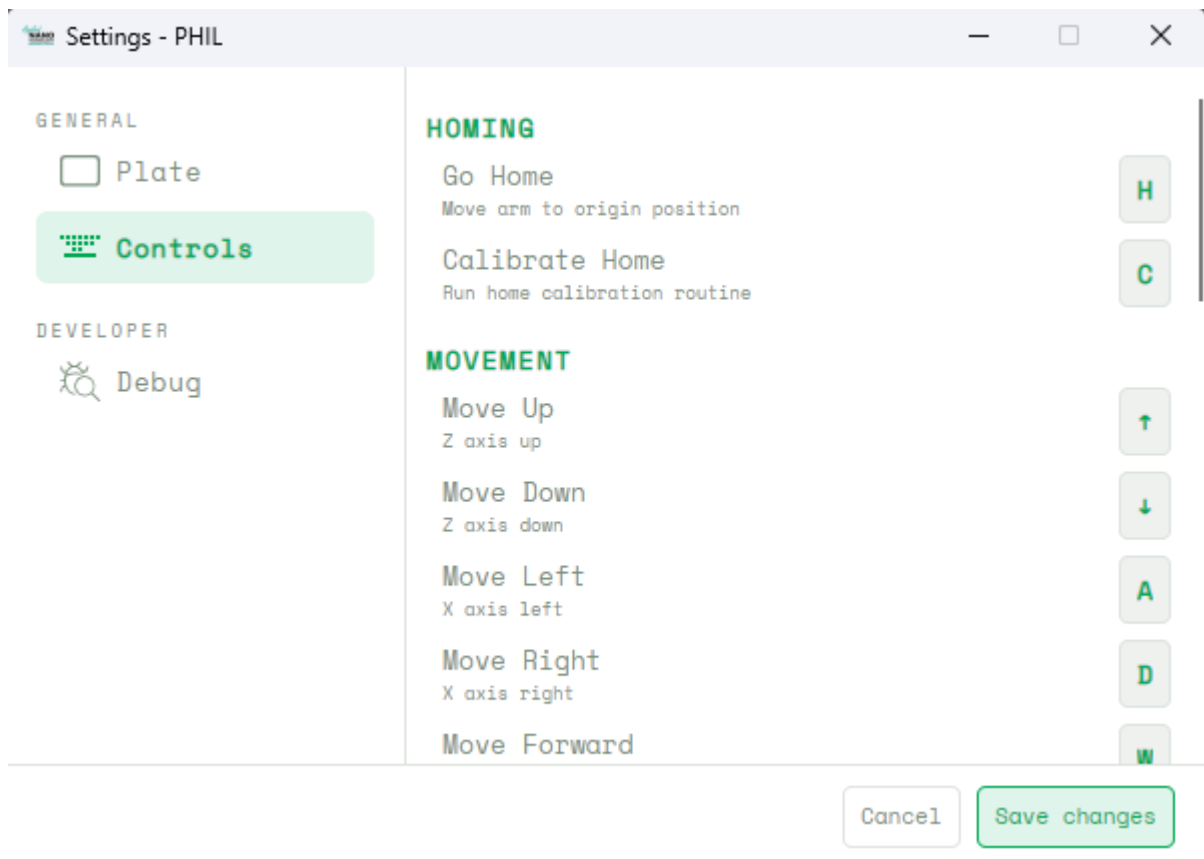
Settings can be opened through the Main Window's Settings button.



Here, we have General section that contains Plate and Controls Settings. Also, we have Developer section, where we have Debug Settings.

Plate Settings is used to change the Plate Type.

Controls Settings gives you the option to see and modify the current Key Bindings.



To modify controls:

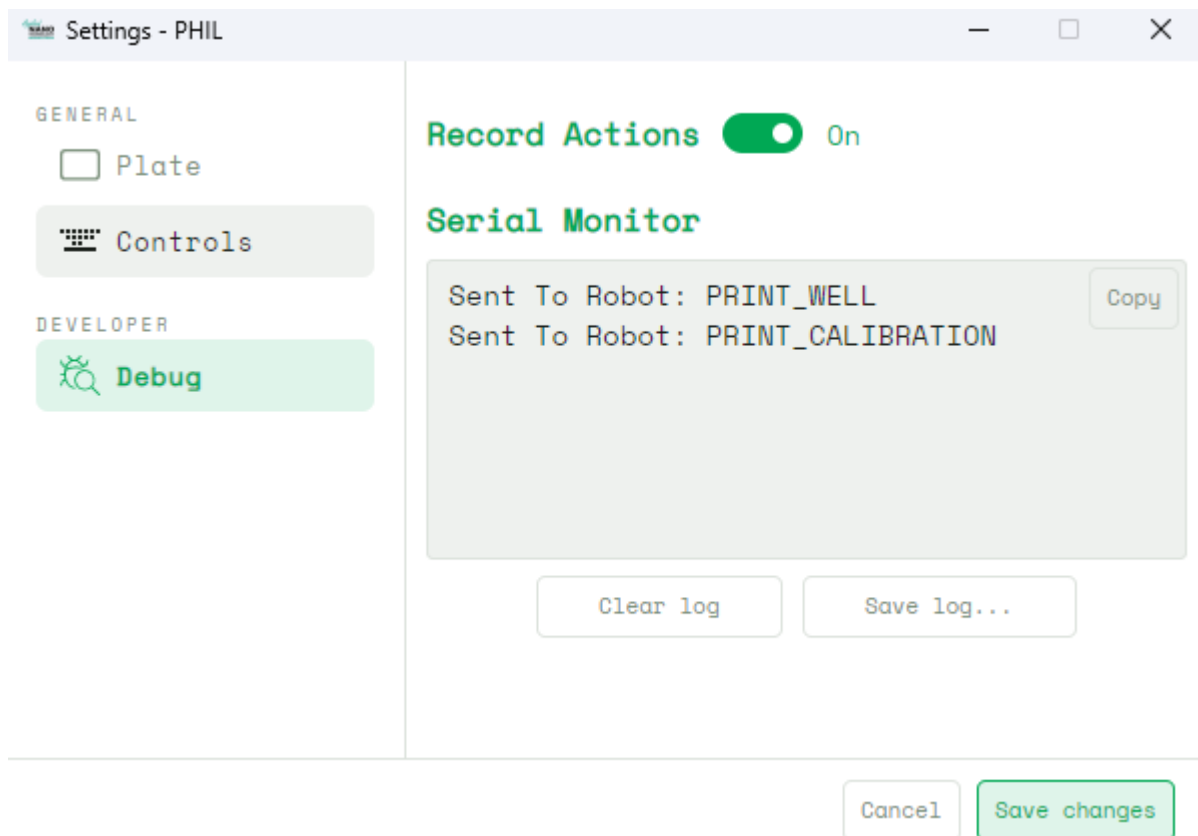
- Click a key binding
- Press new key
- Press Escape to cancel
- Combinations with Shift, Control or Alt are also possible.



In Debug Settings we have the Serial Monitor and Record Actions.

Serial Monitor is used to see the communication between the GUI and PHIL since GUI startup. Log can be copied to keyboard, cleared or saved to a file.

Record Actions is used to record Action execution. If this option is active and if the laptop is connected to PHIL, Action execution will be recorded if a camera is available.

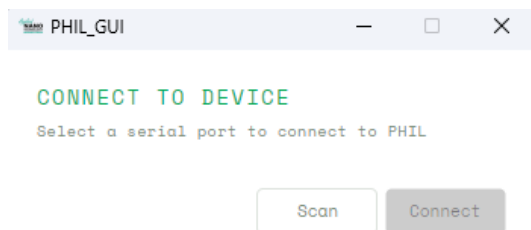


2 Important Notes

- Calibration is required before using wells or actions
- Always verify pipette position before running actions
- Use Emergency Stop if unexpected movement occurs
- Ensure liquid setup is ready before pump operations

3 Getting Started

1. Open the application



2. On the connection screen click Scan to find available ports

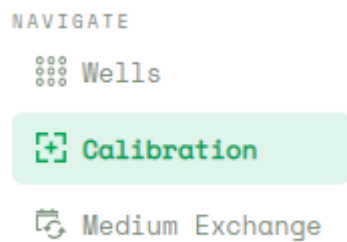


3. Select the correct port (See QA 1 if stuck)
4. Click Connect
5. Wait for Connect and GUI Initialization.

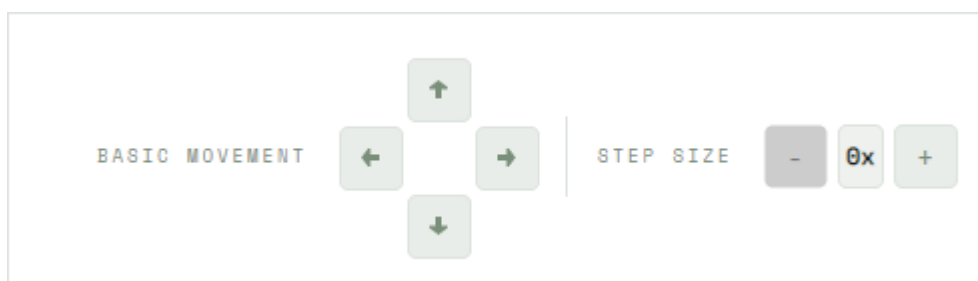
Once connected, the Main Window will open

3.1 Calibrate the System

Go to the Calibration View



1. Click a well (e.g. A1, C5, etc.)
2. On the right sidebar, a popup screen opened. Click on “Go To Well”.
3. The robot moves to an approximate position
4. Use controls or keyboard to adjust



5. Place pipette in center of well
6. Click Record calibration point

RECORD Record position Solve

RECORD Record position Solve

RECORD Record position Solve

7. Repeat for multiple wells (at least 10)
8. In case you recorded a bad point, see Chapter 1.3.
9. Click Solve

3.2 Test Wells

Go to Wells View

NAVIGATE

- Wells
- Calibration
- Medium Exchange

NAVIGATE

- Wells
- Calibration
- Medium Exchange

NAVIGATE

- Wells
- Calibration
- Medium Exchange

NAVIGATE

- Wells
- Calibration
- Medium Exchange

1. Click any well → robot moves there
2. Verify your calibrated but also calculated wells.
3. If one or more wells are off, go back to calibration view and correct.

3.3 Setup Pumps

Open Pump Control popup

Use this to prime tubes or test pump output.

Pump: P1

Volume: 20 μ L

Buttons: Aspirate, Dispense, Prime, Stop

Close

You can:

- Select pump

- Set volume (μL)
- Choose: Aspirate/Dispense

Use Priming to continuously dispense liquid and Stop to manually stop priming.

3.4 Automate with Actions

Go to Medium Schedule View

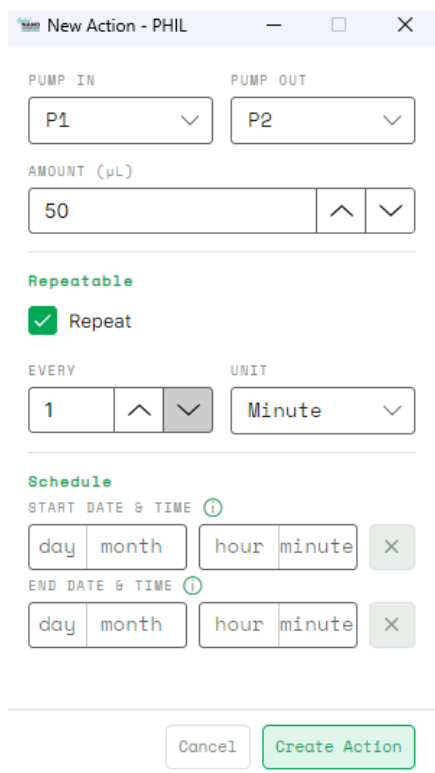
NAVIGATE

 Wells

 Calibration

 Medium Exchange

1. Click Create Action



New Action - PHIL

PUMP IN: P1 PUMP OUT: P2

AMOUNT (μL): 50

Repeatable: ☒ Repeat

EVERY: 1 UNIT: Minute

Schedule

START DATE & TIME: day month hour minute X

END DATE & TIME: day month hour minute X

Cancel Create Action

2. Configure your Action (see Chapter 1.6)
3. Select Channel(s)/Well(s)
4. Attach your Action(s) on the right.

4 Questions and Answers

4.1 How do I know which port to connect to?

Open the GUI without PHIL connected to the PC, scan and see the available ports. Now, close the GUI, connect PHIL and open & scan the ports again. The new port is PHIL.

4.2 If I switch between Plate Types, do I lose my Actions?

No, all Actions are saved in PHIL's memory. The GUI filters and showcases these Actions based on the selected plate type. Actions remain unchanged and they are still attached to your Well(s) or Channel(s). Only the Actions that correspond with the current Plate Type will execute.

4.3 When do I have to Calibrate Home?

If you moved PHIL's arms using your fingers or the pipette hit a wall you must calibrate home again.